

RRSS

System-Wide Requirements Specification

1. Introduction

This document serves as the System Wide Requirements Specification (SWRS) for the ShopSmart project and outlines the key functional requirements that control the operation and behavior of the system.

The primary purpose of this document is to create a common understanding of the system-wide functional requirements necessary for the successful development and implementation of the ShopSmart website. By defining these requirements, stakeholders and developers can align their efforts and ensure that the final product meets the needs and expectations of its users.

2. System-Wide Functional Requirements

ShopSmart will provide the following system-wide functionalities that are essential for its operation and user experience.

2.1 Authentication and Authorization

Requirement 2.1.1: The system will allow users to register for accounts by providing basic information (username, email, password).

Requirement 2.1.2: The system will authenticate users through a secure login process using their registered information (username, email, password).

Requirement 2.1.3: Authorization levels will be implemented to restrict access to certain features and functionalities based on user roles (e.g., admin, merchant, customer).

2.2 Security

Requirement 2.2.1: The system will implement secure password hashing and storage mechanisms to protect user information.

Requirement 2.2.2: The system will restrict unauthorized access to sensitive information (user data, product details) based on user roles (admin, merchant, customer).

2.3 Auditing

Requirement 2.3.1: The system will maintain an audit log that tracks user activities, including login attempts, product listings, reviews submitted, and forum posts created.

2.4 Product Management

Requirement 2.4.1: Merchants will have the ability to add, edit, and remove products from the platform.

Requirement 2.4.2: Product management functionalities will support categorization, tagging, and organization of products to enhance browsing and searchability.

2.5 Review and Rating System

Requirement 2.5.1: Customers will be able to submit reviews and ratings for products, including providing feedback and commentary on their experience.

Requirement 2.5.2: The system will implement mechanisms for moderating reviews and rating.

2.6 Discussion Forum

Requirement 2.6.1: A public discussion forum will be integrated into the platform to help user engagement and community interaction.

Requirement 2.6.2: Forum functionalities will include creating new discussion threads, replying to existing threads, and categorizing discussions by topic or product.

2.7 Search and Filtering

Requirement 2.7.1: The system will offer a search function for users to find products, forum threads, or other information by keyword.

Requirement 2.7.2: The system will allow users to filter product search results based on various criteria, such as category, rating, etc.

2.8 ChatBot

Requirement 2.8.1: The system will feature a ChatBot to assist users with various inquiries, such as product recommendations, FAQ, product performance, support requests, etc.

Requirement 2.8.2: The ChatBot will utilize the transformer architecture using an attention mechanism to understand user input and provide relevant responses in real-time.

3. System Qualities

URPS stands for Usability, Reliability, Performance, and Supportability. These attributes represent essential non-functional requirements that define the overall user experience and system effectiveness. Specifying these qualities ensures the RRSS is not only functional but also meets user expectations in terms of ease of use, dependability, speed, and maintainability. This approach helps ensure a well-rounded system design that considers not just what the system does, but also how well it does it.

3.1 Usability

The system should be easy to navigate for users with varying levels of technical expertise. Users should be able to complete core tasks (writing reviews, finding recommendations and exploring the community) within a limited number of steps. The user interface should be clear, consistent, and visually appealing with its pre-defined identity.

3.2 Reliability

The system should be available for use with minimal downtime due to internal crashes. User data and reviews should be stored securely and protected from loss. The system should be able to handle a reasonable amount of concurrent users without significant performance degradation. Finally, ChatBot feature should produce meaningful response all the time without having a hallucination.

3.3 Performance

Page load times should be within an acceptable range, preferably under 2 seconds. Search queries and recommendations should be generated quickly, preferably under a second. The system should be able to handle at least 100 requests per second without significant delays.

3.4 Supportability

The system should be well-documented with clear code and user guides. Tools and processes should be available for troubleshooting and diagnosing problems. Also, the system architecture should allow for modular upgrades and bug fixes.

4. System Interfaces

4.1 User Interfaces

The user interface (UI) of any software system plays a crucial role in shaping the user experience and determining the overall usability and accessibility of the platform. In this section, we outline the requirements and specifications pertaining to the UI design and functionality of our system. The UI design encompasses various aspects, including the look and feel, layout, navigation, consistency, and user personalization. By defining clear guidelines and objectives for the UI, we aim to create an intuitive, visually appealing, and user-friendly interface that enhances user engagement and satisfaction. Let's delve into the specifics of each aspect to ensure a comprehensive understanding and implementation of our UI requirements.

4.1.1 Look & Feel

Users will primarily interact with the system using mouse pointers. Key features of the platform will be easily accessible, typically requiring only a few clicks from the homepage. Navigation will be facilitated through a sticky navigation bar located at the top of each page, ensuring quick access to essential features regardless of scrolling position. While priority will be given to desktop displays, mobile responsiveness is optional but can be implemented as an additional feature.

4.1.2 Layout and Navigation Requirements

Users will primarily interact with the system using mouse pointers. Key features of the platform will be easily accessible, typically requiring only a few clicks from the homepage. Navigation will be facilitated through a sticky navigation bar located at the top of each page, ensuring quick access to essential features regardless of scrolling position. While priority will be given to desktop displays, mobile responsiveness is optional but can be implemented as an additional feature.

4.1.3 Consistency

Consistency in the user interface design is paramount to provide users with a predictable and intuitive browsing experience. Therefore, the layout pattern across all pages of the platform will remain consistent. This consistency extends to terminology usage, minimizing ambiguity and complexity in user interactions. Users can expect uniformity in navigation controls, screen area sizes and shapes, as well as data entry and presentation methods throughout the system.

4.1.4 User Personalization & Customization Requirements

The displayed content will be personalized to each user based on their attributes such as name and profile picture. Additionally, users will have the ability to customize their profiles, including changing their profile picture and updating contact information. This level of personalization and customization aims to enhance user engagement and satisfaction with the platform.

4.2 Interfaces to External Systems or Devices

4.2.1 Software Interfaces

4.2.1.1 ChatBot Interface

The software interface to be established will facilitate communication between our main system and the external chatbot service responsible for providing user support. This interface will define the protocols, data formats, and communication methods required for seamless interaction between the two systems. The interface will allow our main system to send user inquiries and receive responses from the external chatbot service, supporting functionalities such as message exchange, query handling, and retrieval of support information.

Communication between the main system and the external chatbot service will be facilitated using standard HTTP(S) protocols, with data exchanged adhering to a predefined JSON format. Integration points within

our main system's architecture will enable seamless invocation of chatbot functionalities and retrieval of responses to user inquiries.

Additionally, the interface must ensure reliability, security, scalability, and effective error handling, with comprehensive documentation provided to facilitate integration and development efforts.

4.2.1.2 SQLite Database Interface

In addition to the interface with the external chatbot service, our main system will also interact with a SQLite database for data storage and retrieval purposes. This interface will facilitate communication between our software application and the SQLite database, enabling efficient management of user data, system configurations, and other relevant information.

The interface will specify protocols and query languages for database interaction, ensuring compatibility with SQLite's relational database management system. Integration points within our main system's architecture will support CRUD (Create, Read, Update, Delete) operations on the SQLite database, allowing for data manipulation and retrieval as required.

Security measures, such as access control and data encryption, will be implemented to safeguard sensitive information stored within the database. Comprehensive documentation detailing database schemas, query structures, and integration guidelines will be provided to facilitate development and maintenance efforts.

4.2.1.3 Recommendation System Interface

In addition to the aforementioned interfaces with the external chatbot service and the SQLite database, our main system will also incorporate an interface with a recommendation system. This recommendation system will be responsible for delivering personalized product recommendations to users based on their preferences, browsing history, and reviews.

The interface will facilitate communication between our software application and the recommendation system, ensuring the efficient transmission of user data and receipt of recommendation results. Protocol specifications and data formats will be defined to govern the exchange of information between the systems. Integration points within our main system's architecture will enable the retrieval and processing of user-related data to generate pertinent recommendations.

Robust security measures, including data encryption and access controls, will be implemented to uphold user privacy and safeguard sensitive information utilized by the recommendation system. Comprehensive documentation elucidating the recommendation algorithms, data sources, and integration protocols will be furnished to facilitate the seamless integration and utilization of the recommendation system within our software application.

4.2.2 Communications Interfaces

Communication interfaces play a crucial role in enabling data exchange and interaction with other systems or devices, whether locally or remotely. These interfaces serve as conduits for transmitting information, commands, and requests between our system and external entities, ensuring seamless integration and interoperability.

Communication interfaces are essential components enabling our system to connect with external systems or devices, facilitating seamless data exchange and interaction. Utilizing communication protocols such as HTTP(S), TCP/IP, and WebSocket ensures reliable and secure transmission of data, while adhering to standardized formats like JSON or XML facilitates interoperability and compatibility. Integration points within our system's architecture enable smooth interaction with external entities, supporting functionalities such as data retrieval, synchronization, and real-time communication. Moreover, ensuring reliability, security, scalability, effective error handling, and comprehensive documentation of the communication interfaces are crucial to facilitate integration, development, and troubleshooting efforts, ultimately ensuring efficient operation and fulfillment of user requirements.

5. Business Rules

5.1 User Validation Rules

5.1.1 Age Restriction

If a user's date of birth is earlier than 18 years from the current date, suspend the user's account.

5.1.2 Name Restriction

If a user's name contains profanity, suspend the user's account.

5.1.3 Profile Image Restriction

If a user's profile contains immoral or illegal content, suspend the user's account.

5.2 Product Rules

5.2.1 Product Legitimacy

If a product is determined to be a fraud, remove the product and suspend the merchant's account.

5.2.2 Product Image

If a product contains immoral or illegal images, remove the product and suspend the merchant's account.

5.2.3 Featured Products

If a product is in the top 10 most viewed products of the month, highlight the product on the main page.

5.3 Community Engagement Rules

5.3.1 Order Rule

If a community topic has a new entry, push it to the top of the topics list.

5.3.2 Entry Rule

If an entry contains inappropriate content, remove the entry and suspend the user's account.

6. System Constraints

6.1 Software Implementation Languages

The project will be mainly implemented in Java™, on top of OpenJDK runtime. Additional languages will also be used for templating engines and stylesheets.

6.2 Third-Party Components and Libraries

The most vital parts of the project will be built on top of the Java Spring Boot Framework. For the frontend part, Thymeleaf templating engine will be used. SQLite will be used as the database.

6.3 Development Tools

To ensure a higher level of code quality, various tools and methods will be employed such as formatters, linters, CI/CD pipelines, unit tests, integration tests.

6.4 Platform Support and Resource Limits

For instance hosts, a server with a POSIX compliant OS with OpenJDK 21 support; 2 vCPUs and 4

gigabytes of RAM is required.

For end users, a computer with a clock speed of at least 1 gigahertz, 2 gigabytes of RAM, 500 megabytes of hard drive space and a recently updated operating system such as Windows 10, MacOS 10.15 or Linux is recommended.

7. System Compliance

7.1 Licensing Requirements

This project is released under the terms of the GNU Affero General Public License version 3. This software is released “as is”, without warranty of any kind, express or implied. In no event shall the authors or copyright holders be liable for any claim, damages or other liability, whether in action of contract, arising from, out of or in connection with the software or the use or other dealings in the software. By using this project, users and hosts of this project hereby agree to the following:

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Additionally, all end users of the software agree to, that they either are the copyright holders or have received permission from the copyright holders of any text or media uploaded or submitted to the instance of this software. They also agree to give an irrevocable license to the instance owners solely for the purpose of sharing the text or media to the users of the instance, up until removal by an administrator or a deletion triggered by user after a period of time governed by relevant laws.

7.2 Legal, Copyright, and Other Notices

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This application depends on:

- OpenJDK which is licensed under GNU General Public License with linking exception,
- Java Spring Boot Framework which is licensed under the Apache License,
- Thymeleaf which is also licensed under the Apache License,
- Mistral 7B v0.1 which is also licensed under the Apache license.

7.3 Applicable Standard

This project follows the following legal standards:

- Turkish Law No: 5651, The Law on the Regulation of Broadcasts via Internet and Prevention of Crimes Committed through Such Broadcasts;
- Turkish Law No: 6698, The Law on Protection of Personal Data;
- EU Regulation 2016/679, General Data Protection Regulation;
- California Civil Code AB-375, California Consumer Privacy Act.

This project also follows the following international standards:

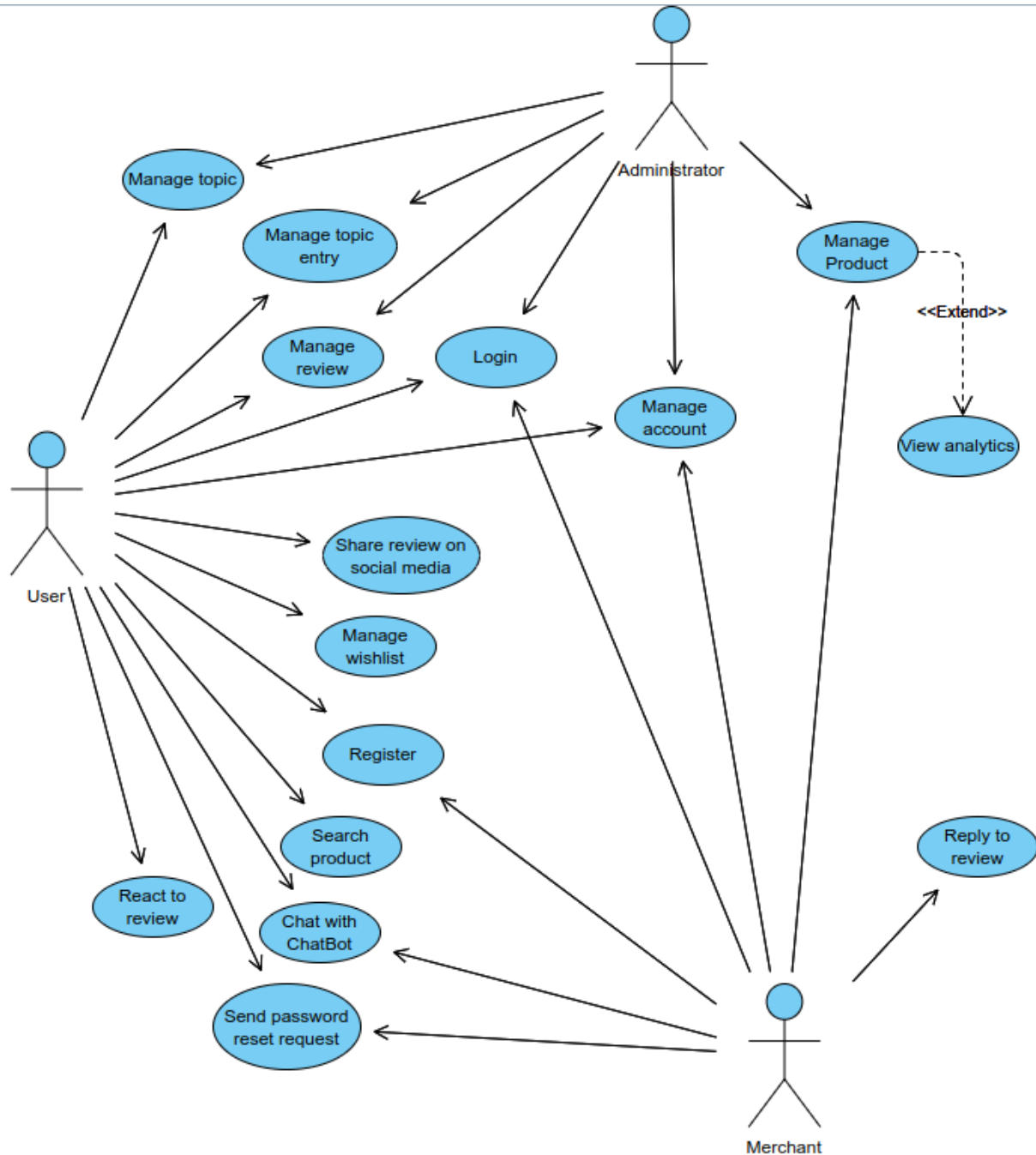
- IEEE RFC 9112, HTTP/1.1;
- IEEE RFC 9113, HTTP/2;
- IEEE Std 1003.1-2017, Portable Operating System Interface Standard;
- WHATWG HTML5 Specification;
- WHATWG CSS3 Specification;

8. System Documentation

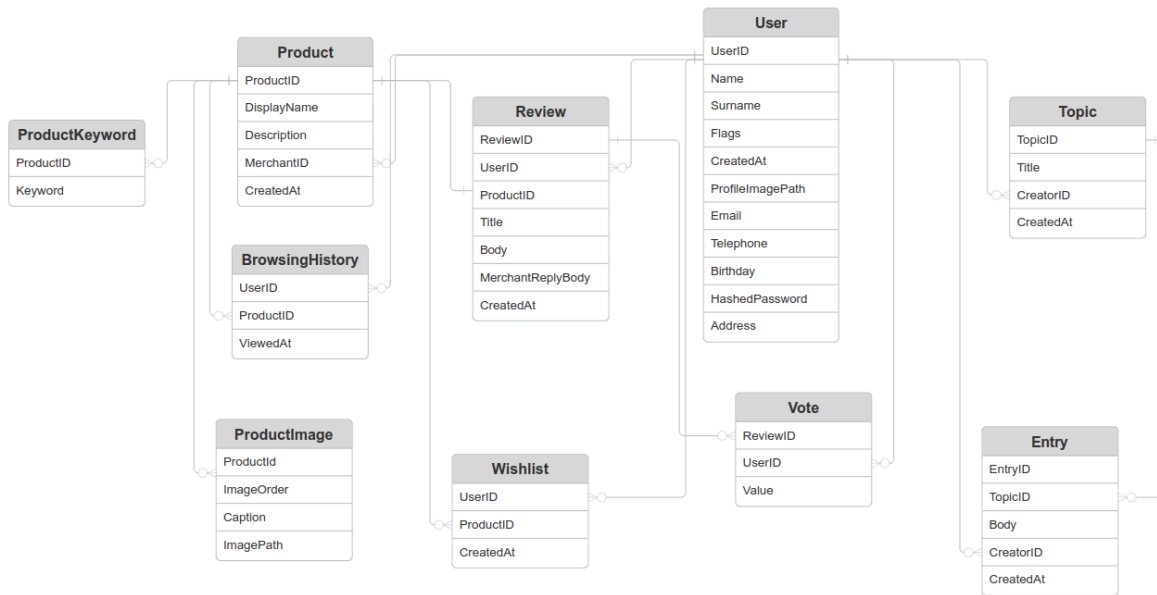
Assistance for using the system for users will primarily be provided by the internal automated assistance system(also referred as chatbot) and an external support team providing support using e-mail. Instance administrators are also expected to be responsive on community forums to assist users.

9. Appendices

9.1 High-Level Use Case Diagram

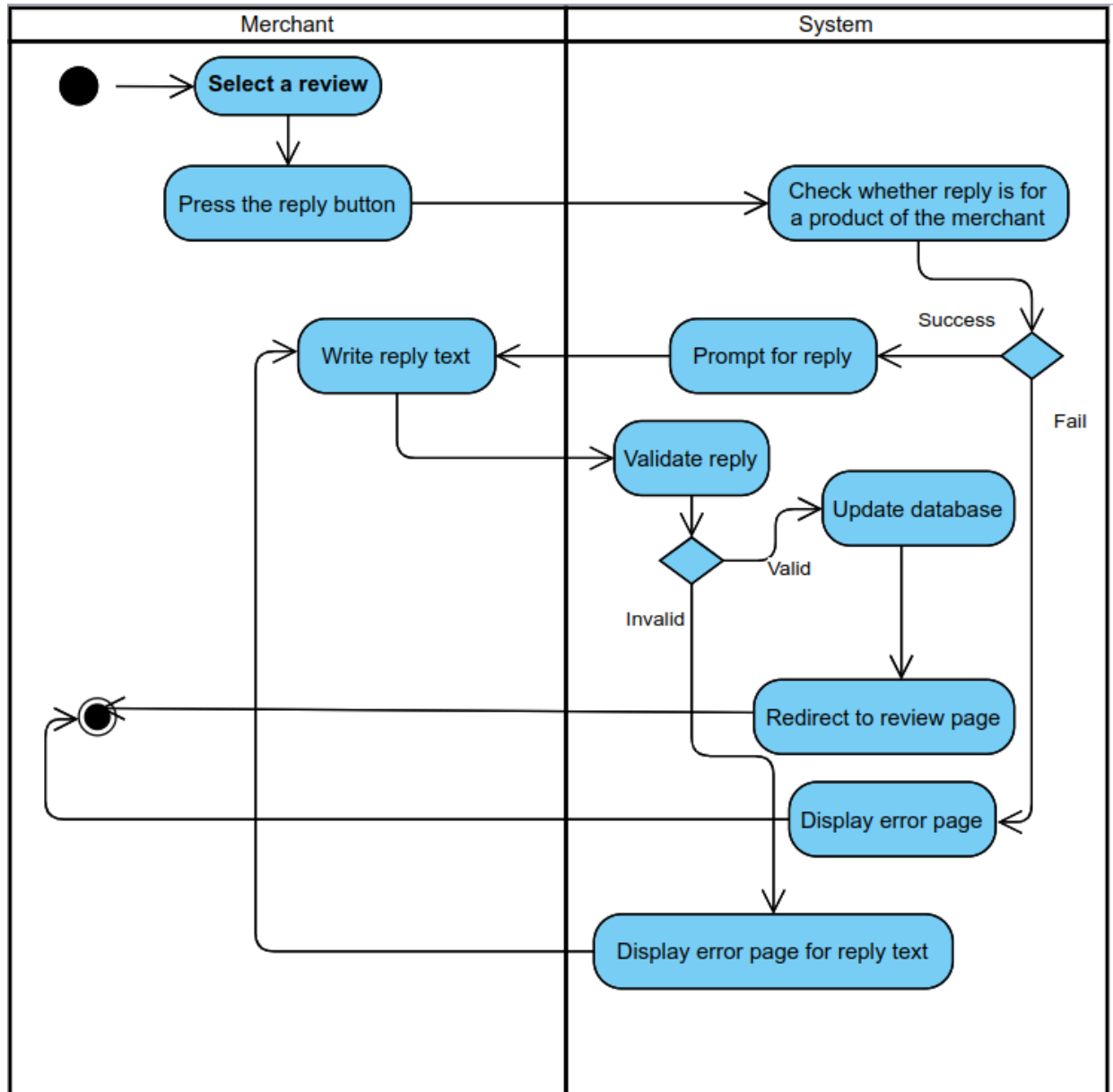


9.2 ER Diagram

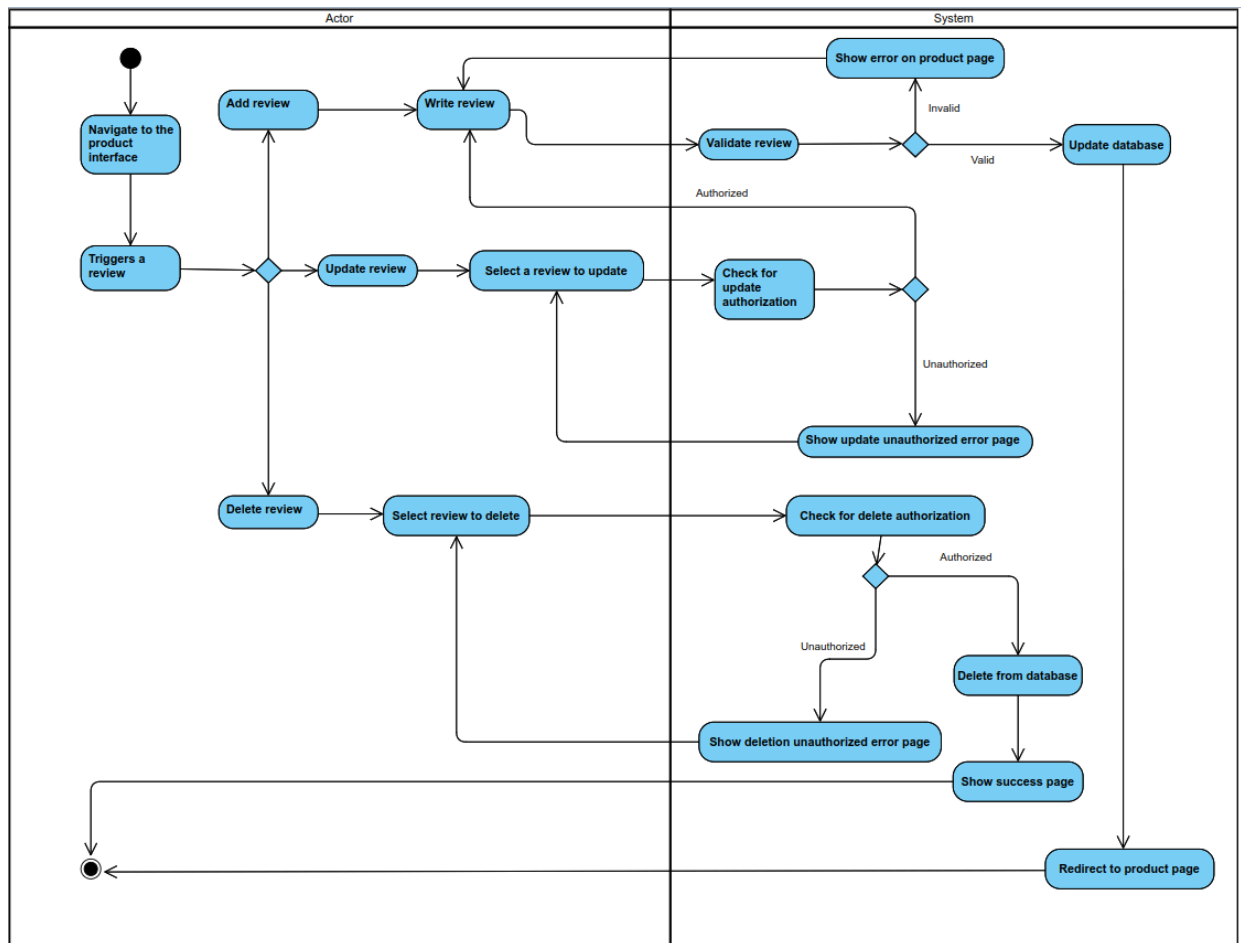


9.3 Activity Diagrams

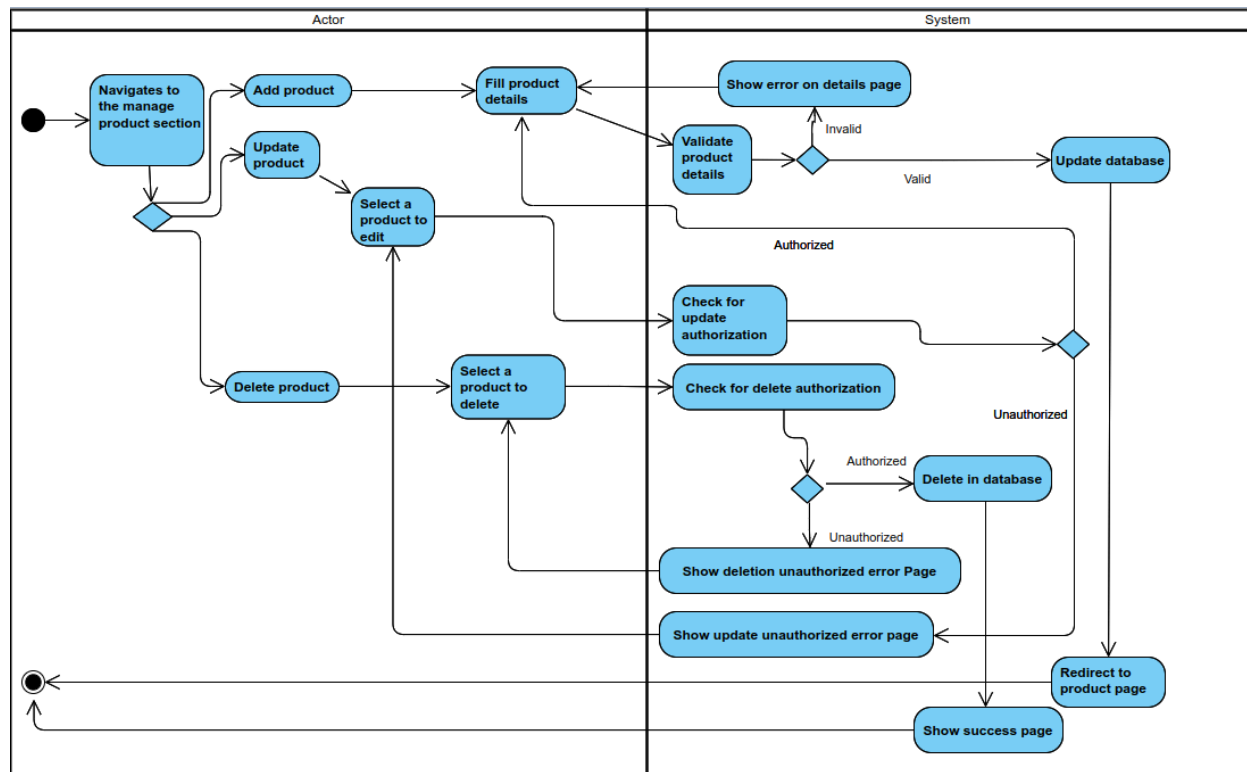
9.3.1 Activity Diagram for "Reply to Review"



9.3.2 Activity Diagram for "Manage Review"



9.3.3 Activity Diagram for "Manage Product"



9.4 Traceability Matrix

	Furkan Şimşekli	İlker Avcı	Mete Uysal	Eyüp Menevşe	Buğra Kağan Acar
High-Level Use Diagram	+				
ER Diagram		+			
Activity Diagrams			+		+
Traceability Table			+		+
System-Wide Requirements Specification	+	+	+		+
GUI Design Document				+	
Use Cases	+	+	+	+	+
Test Cases	+	+	+	+	+
Time/Duration	12 hours	9 hours	10 hours	8 hours	11 hours